DCSN: Deep Compressed Sensing Network for Efficient Hyperspectral Data Transmission of Miniaturized Satellite

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***Abstract*— Requirements of compressed sensing techniques targeted at miniaturized hyperspectral satellite applications include lightweight onboard hardware, high-speed sensing, low sampling rate for compressing the massive volume of typical hyperspectral data, and noise robustness for reliable data transmission to the ground station. We achieve all these aims via deep learning, and neural networks resulted from which can be implemented on-chip, thereby allowing light hardware implementation. Our neural networks were trained from small-scaled data, but, even so, the resulting encoder achieves a very low sampling rate and very high speed. Unlike typical network training, the input–output pairs are not square but stripe-like images, partly because compressed acquisition does not allow performing compression after obtaining complete data cube and partly because stripe-like acquisition well matches the popular pushbroom hyperspectral sensing schemes. Even with such hard restriction caused by nontraditional training, the resulting decoder still reconstructs the image with high accuracy. To match the requirement of pushbroom sensing, a lightweight encoder is proposed to compress the stripe-like images immediately. Meanwhile, multiscale feature fusion block (MFB) and aggregation (MFA) modules are proposed to form our decoder for enhancing the feature representation of the compressed acquisitions. Furthermore, we achieve joint spatial/spectral super-resolution (SR) progressively, ensuring accurate hyperspectral reconstruction via a low-rank-driven decoder. The encoder and decoder are trained in an end-to-end manner, where noise robustness is forced during the training stage. Comprehensive experiments demonstrate the superiority of the proposed hyperspectral compressed sensing method, as well as its one-shot transfer learning (OTL)-based extension, both quantitatively and qualitatively.**

***Index Terms*— Compressed sensing, deep learning (DL), hyperspectral image, miniaturized satellite, multiscale feature fusion.**

I. Introduction

HYPERSPECTRAL remote sensing is critical for numerous science and engineering applications, including environmental monitoring, precision agriculture, land mapping/classification, and planetary exploration [1]–[5]. However, hyperspectral imagery (HSI) typically involves hundreds of spectral bands, ranging from visible to invisible spectral regions (see Fig. 2), leading to a huge volume in typical HSI. Directly transmitting HSI data acquired on satellite to the ground station would not be efficient and may even be infeasible due to the limited bandwidth of the communication channel between the satellite and the ground station. This motivated a series of research works on hyperspectral compressed sensing (HCS); see, e.g., the benchmark spectral compressive acquisition (SpeCA) [6] and the references therein.

On the other hand, we would like to mention some outstanding principal component analysis (PCA)-based hyperspectral compression methods, including the "PCA+JEPG2000" [7] and "PCA+X264" [8], though hyperspectral compression and HCS are two completely different topics. The two PCA-based methods are quite effective in reducing the size of HSI data, but they ignore the trend of the hyperspectral pushbroom sensing system in the target application, which will be carefully considered in this article for customizing a deep network structure for effective strip-like HCS, as will be detailed later. Also, HCS methods should simultaneously perform compression and sensing, while conventional compression methods are allowed to use the information of the entire HSI after the sensing stage is completed, implying that HCS is more challenging. Furthermore, most hyperspectral compression methods were developed for onground applications, hence ignoring the noise-robustness issue during the transmission; for example, the entropy coding may not be well decoded from the noise-corrupted bit stream of the compressed HSI if there is no redundant coding or error concealment mechanism, preventing the adoption of the conventional compression methods [7], [8] in our target application. For the miniaturized satellite application, there are inevitable noise corruptions when transmitting the compressed acquisitions to the ground station, so developing new HCS techniques is needed. Furthermore, the

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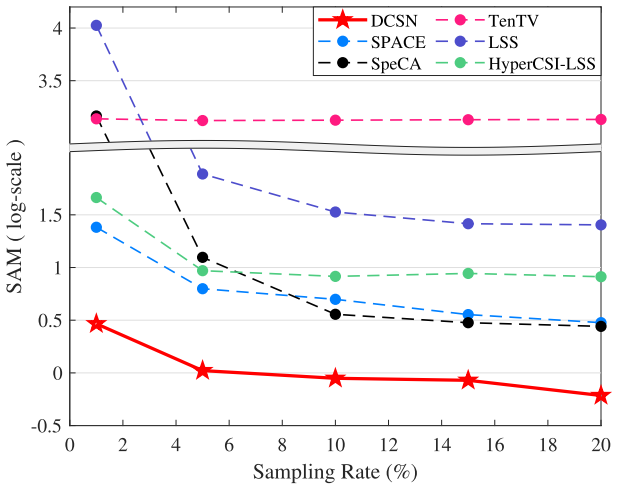


Fig. 1. Performance comparison between the proposed DCSN and state-of-the-art HCS methods in terms of SAM (log-scale), a key index for measuring spectral distortion in the reconstructed HSI.

onboard computing/storage resources are also limited, making most existing HCS methods not suitable for miniaturized satellites (e.g., CubeSat). Different from ground applications, the HCS strategy on satellite requires lightweight hardware and computationally light measurement strategy. Therefore, advancing hardware-friendly HCS technology based on deep learning (DL) is appealing as motivated by recent advances in the networks-on-chip (NoC) technology [9], [10]. Before introducing DL-related techniques, we review some state-of-the-art compressed sensing and reconstruction methods.

One category of HCS methods employs sparse recovery based on the ℓ_1-norm regularization or total-variation (TV) regularization [11], [12]. A benchmark Tensor TV (TenTV) method was proposed in [12]. TenTV exploits the Tucker decomposition to model the global spatial–spectral correlation of HSI and characterizes local smoothness by designing a weighted TV regularizer. It is very interesting to notice some HCS works that connect the HCS problem with another important blind signal processing problem, known as hyperspectral unmixing (HU) [13], [14]; see, e.g., the locally similar sparsity-based HU compressive sensing (LSSHUCS, or LSS for brief) method [15]. Though the LSS theory is elegant and appealing, we found that its reconstruction is not so robust. After deeply investigating the experimental results, we found that the default redundant library embedded in LSS is not a good one for HU. Therefore, we employ the hyperplane-based Craig simplex identification (HyperCSI [14]) algorithm to obtain a scene-adapted library for LSS, and as it turned out, the reconstruction performance of HyperCSI-LSS is significantly improved, as detailed in Section III.

Just like many DL theories, reducing the number of parameters is essential for having stable performance. Low-rank representation of a hyperspectral data matrix is also regarded as a key to reduce the parameters, having also been applied for HCS [16]. For example, SpeCA [6] exploits the fact that hyperspectral pixels often belong to a low-dimensional subspace, thereby yielding an encoder that just computes random projections (RPs) targeted at recording the required information in matrices reconstruction. Very recently, a novel spatial/spectral compressed encoder (SPACE) has also been introduced [17]. Due to the low rankness in typical HSI, SPACE judiciously records the spatial information and spectral information of the HSI in the coding stage, and in the decoding stage, SPACE transforms the decoding problem into an image fusion problem for fast decoding. More specifically, SPACE transforms the acquired spatial information into a high-spatial-resolution panchromatic image and spectral information into a high-spectral-resolution HSI, followed by super-resolving the HSI via the aid of the panchromatic image. Such super-resolution (SR) problem is exactly the well-known panchromatic sharpening problem in satellite remote sensing and allows the adoption of existing convex optimization solver [18] for simple and fast decoding. This motivates us to think the following question: *considering that SR is able to reconstruct full HSI from much fewer acquisitions and recent state-of-the-art SR techniques are more oriented from the DL advances, can we achieve HCS via SR-based deep neural networks?*

The answer is affirmative; to see it, we first introduce some related backgrounds hereinafter. With the rapid growth of DL, several DL-based schemes for HSI applications have been proposed in recent years. One significant advantage from the DL-based approach is that highly effective feature representation can be learned, leading to boost the performance in various computer vision applications, such as image SR [19]–[21], compression [22], denoising [23], object detection [24], and semantic segmentation [25]. To reduce both spectral and spatial dimensionality, autoencoder (AE) [26] is widely used to learn the compact feature representation of the training data. In HCS applications, AE is one of the candidates that can achieve simultaneous spatial and spectral information compressions. However, traditional AE is constructed by stacking several fully connected layers in which the feature representation ability is ineffective for 2-D data (i.e., the general image or hyperspectral data). In [27], a convolutional AE (CAE) is presented to better capture the compact features embedded in images. However, most CAEs are used to extract the image features, rather than generating high-dimensional data from the compact features. As it is hard to reconstruct visual-pleased results from the low-dimensional data by merely adopting CAE, several methods, such as generative adversarial networks (GANs) [28], were proposed recently to resolve this shortcoming. Several advanced techniques were proposed in [19] and [20], showing the excellent performance, while they can only perform SR along the spatial dimension. Besides, as reported in [19] and [20], the common objective quality of the reconstructed image will be lower than that of conventional convolutional neural network (CNN)-based SR techniques. In our proposed HCS method, the hyperspectral data must be accurately recovered (from the compressed acquisition) also along the spectral dimension, as the key value of HSI lies within material identification that, however, requires preserving the spectral shape of the material signatures. As a result, GAN-based SR is not considered for hyperspectral CS applications.

Different from traditional image SR techniques [19]–[21], the number of channels in hyperspectral data is significantly larger than that of the generic RGB image. To effectively

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compress the hyperspectral data, it is necessary to reduce the spatial resolution and compress the channel information (i.e., spectra). Therefore, the compressed data should be recovered in both spatial and spectral resolutions. The conventional image/HSI SR methods only focus on recovering the spatial resolution. A novel HSI SR technique was proposed in [29]. A modified and straightforward Residual Network (ResNet) [30] is used for upsampling the low-resolution (LR) HSI in the spatial domain. The experimental results verified that such an HSI-SR method is effective. In [31], a hybrid HSI-SR method was designed by combining collaborative nonnegative matrix factorization and a pretrained SR-CNN. By using transfer learning, the SR-CNN can be learned on the general images so that it is unnecessary to collect a large-scale data set of HSI. However, it is a high cost on hardware design because the extra memory and storage space are required in the satellite while sensing hyperspectral data to form complete 2-D HSI.

In [32], a novel SR network for hyperspectral data is proposed based on the optimized RGB guidance. The high-resolution (HR) RGB image from hyperspectral data is used as the regularizer. Therefore, the LR hyperspectral data can be effectively super-resolved. However, the large number of hyperspectral channels/bands is still not well resolved, and the spectral resolution is not considered in their SR architecture, causing inefficient satellite transmissions. As a result, the existing HSI-SR approaches concentrate on the spatial SR, ignoring the spectral dimension, and, hence, are not able to achieve a low sampling rate in the target HCS application. To the best of our knowledge, though joint learning across all spectral bands/channels has been reported to be critical in multichannel SR [33, Sec. IV-C], there is no joint hyperspectral compression SR network architecture for both spectral and spatial domains.

This article presents a deep compressed sensing network (DCSN), which is suitable for miniaturized satellite application due to the aforementioned NoC advances, the high-speed encoder (discussed below), and the low sampling rate achieved by DCSN [i.e., $\approx 1\%$ (see Fig. 1)]. Since the computational power in the miniaturized satellite is limited, the computational complexity of the encoder (say, compression process) should be minimized.

Different from compression, compressed sensing has to perform data compression while sensing. In this regard, we need to specify the sensing scheme before designing DCSN. Since the fast movement of the sensing hardware required by the whiskbroom sensing easily causes aging of the devices, the design of DCSN will be based on the pushbroom sensing [34]. Specifically, the pushbroom system scans the image via strip-by-strip acquisition (i.e., pushbroom scanning), so, considering the limited onboard storage resources, DCSN needs to compress strip-like HSI immediately, as illustrated in Fig. 2. To minimize the run time and memory usage, the input of the proposed DCSN should be designed to be a stripe-like image acquired by the pushbroom system, posing one challenge in the network design. Another challenge comes from real-time data compression/transmission required by satellite applications, meaning that the encoder part of DCSN should be computationally very cheap. Therefore, the proposed encoder consists of only three convolutional layers for nonlinearly compressing the stripe-like hyperspectral data. However, such a greatly simplified encoder may not learn powerful and robust feature representation to be recorded in the compressed acquisition, making the decoder design of DCSN highly challenging in accurately reconstructing the HSI data from the compressed one. To overcome this shortcoming, we propose a multiscale feature fusion block (MFB) and multiscale feature aggregation (MFA) unit by stacking three MFBs to further improve the feature representation of the compressed hyperspectral features.

In order to address the issue of super-resolving spectral and spatial resolution jointly, a highly effective network architecture is proposed for decoding and super-resolving the compressed hyperspectral acquisition on the receiver side of the ground station. Inspired by the recent advances in convex optimization-based HCS [17], where a low-rank representation of the HSI $\boldsymbol{x} = \boldsymbol{\Psi s}$ is employed to reduce the number of parameters by one order of magnitude, we design DCSN to take this advantage. The method in [17] first puts most energies on solving a convex SR criterion, i.e., [17, eq. (3)], to yield a high-spatial-resolution $\boldsymbol{s}$, referred to as eigenmaps, which are then left-multiplied by a basis matrix $\boldsymbol{\Psi}$ to spectrally upsample the $N \approx 10$ eigenmaps to $M \approx 100$ spectral bands in the reconstructed HSI. Then, though DL is nonlinear, the above low rankness in linear algebra can still be trickily fused into the DCSN architecture. Simply speaking, in the first layers of the proposed decoder, most energy is concentrated on first upsampling the spatial resolution via joint adoption of the proposed MFA and the transposed convolution operation. Once the high-spatial-resolution feature maps are available, they are spectrally upsampled by one additional convolutional layer, which enhances the spectral resolution of the reconstructed hyperspectral data; note that if the spectral-upsampling layer is set from the beginning, the number of parameters and the memory requirement in DCSN will be significantly much larger because the number of channels of the convolutional layer becomes large. In this way, the proposed DCSN can efficiently compress/encode the stripe-like hyperspectral data, as well as effectively decode the compressed signal on the ground. Finally, the primary contribution of this article is threefold:

1) To the best of our knowledge, DCSN, particularly designed for the miniaturized satellite application, is the first work introducing DL into joint HCS/SR. The joint HCS/SR architecture involves a highly efficient encoder and a low-rank-driven decoder for joint spectral/spatial compression and enhancement, and the encoder/decoder networks are trained in an end-to-end [35] manner.
2) Our lightweight encoder is designed to be computationally very cheap and to match the trend of pushbroom hyperspectral scanning system, allowing the real-time encoding of stripe-like HSI data.
3) The noise robustness of the proposed decoder in DCSN has been achieved by emphasizing the feature representation in the decoding stage, while the number of parameters involved in DCSN is reduced by a low-rank-driven design (i.e., spatial SR first, followed by spectral

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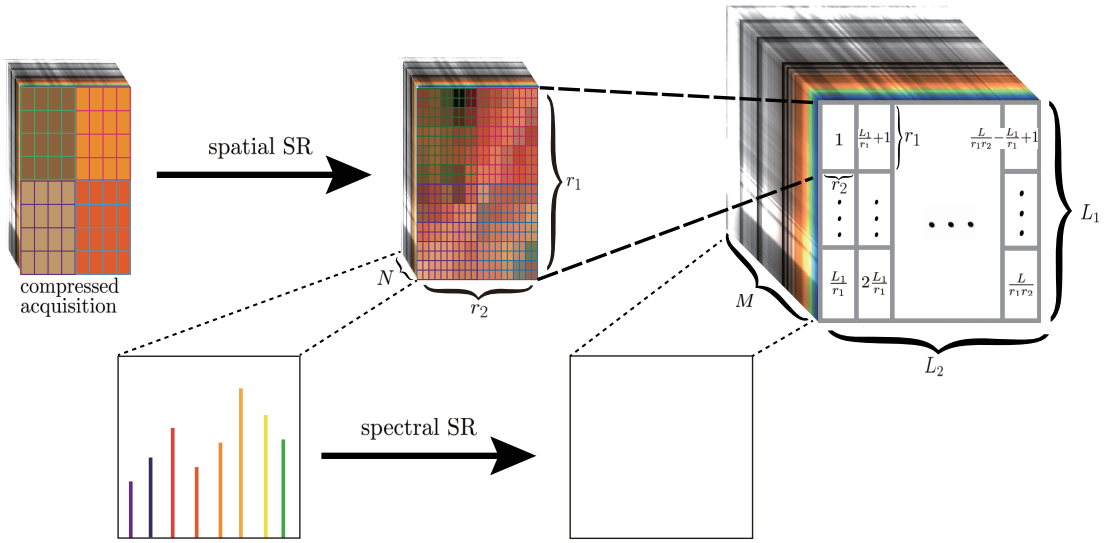


Fig. 2. Conceptual illustration of low-rank-driven decoder architecture. Simply speaking, the decoder in DCSN jointly performs spatial and spectral SR, and the ordering of layers is inspired by the spirit of low-rank modeling in recent HCS advances for reducing the number of parameters.

SR), thereby ensuring high-accuracy HSI reconstruction even with very low sampling rate (e.g., 1%).

The rest of this article is organized as follows. Section II presents the proposed DCSN architecture design. In Section III, superiority of DCSN over benchmark methods is experimentally demonstrated. Finally, conclusions are drawn in Section IV.

II. Proposed Method

Unlike the traditional image, the hyperspectral data typically have hundreds of spectral bands for each pixel, as shown in Fig. 2. Each dimension of the hyperspectral data (i.e., spatial information and spectral information) should be well compressed and reconstructed, allowing effective data transmission between the space and the ground. To meet this requirement, we propose a DL-based approach to jointly compress spectral information and spatial information and simultaneously super-resolve/decode the compressed acquisition along both dimensions.

A. Overview of the Proposed DCSN

The idea is illustrated in Fig. 3, where one can find that the proposed DCSN is composed of a lightweight encoder and an effective decoder for joint SR/HCS. The overall flow is to compressively sense the hyperspectral data by our encoder in real time and then to decode the compressed signal by our decoder. For real-time encoding and transmission, a lightweight encoder consisting of just three convolutional layers is designed. Each convolutional layer only consists of the convolution operation and the LeakyReLU activation function. The dimensions of the three convolutional layers are 128, 64, and k_{ch}, respectively, where k_{ch} is the number of the last convolutional layer determined by the sampling rate. It is worth noting that the storage devices are no longer required onboard the miniaturized satellite under the stripe-like sensing scheme (see the left-hand side in Fig. 3). Afterward, the compressed stripe-like hyperspectral data will be transmitted to the receiver on the ground station without requiring ample buffer space. However, effective feature representation is hard to be learned through the highly simplified three-layer encoder network architecture. Therefore, a highly effective joint SR/HCS deep neural network is needed. To this end, in our decoder, we propose an MFA to effectively characterize the multiscale feature information from the compressed signal. Then, several MFAs are stacked to further enhance the ability of feature representation, followed by using the upsampling layers to upscale the spatial resolution of the compressed data. Finally, two additional convolutional layers are adopted to super-resolve the spectral resolution, corresponding to the right part of Fig. 3.

This ordering of layers is inspired by the recent advances in HCS [17]; specifically, a low-rank model is employed in [17] to reduce the number of parameters by around one order of magnitude, allowing first concentrating on decoding the spatial details, followed by left-multiplying a basis matrix to decoding the spectral information of the reconstructed HSI. In this way, the proposed SR-aware decoder can reconstruct the hyperspectral data more accurately and efficiently. The details of each component of the proposed method are described in the following subsections.

B. Lightweight Encoder

Considering the limitations of computational power, memory resources, and the pushbroom sensing scheme on miniaturized satellites, we are facing the following three challenges.

1) Since the miniaturized satellite acquires HSI strip-by-strip, the strip-like data should be immediately compressed while sensing. However, most deep neural networks and the involved convolution kernels are designed to have nonstripe-like input–output pair images, making them not directly applicable in practice.

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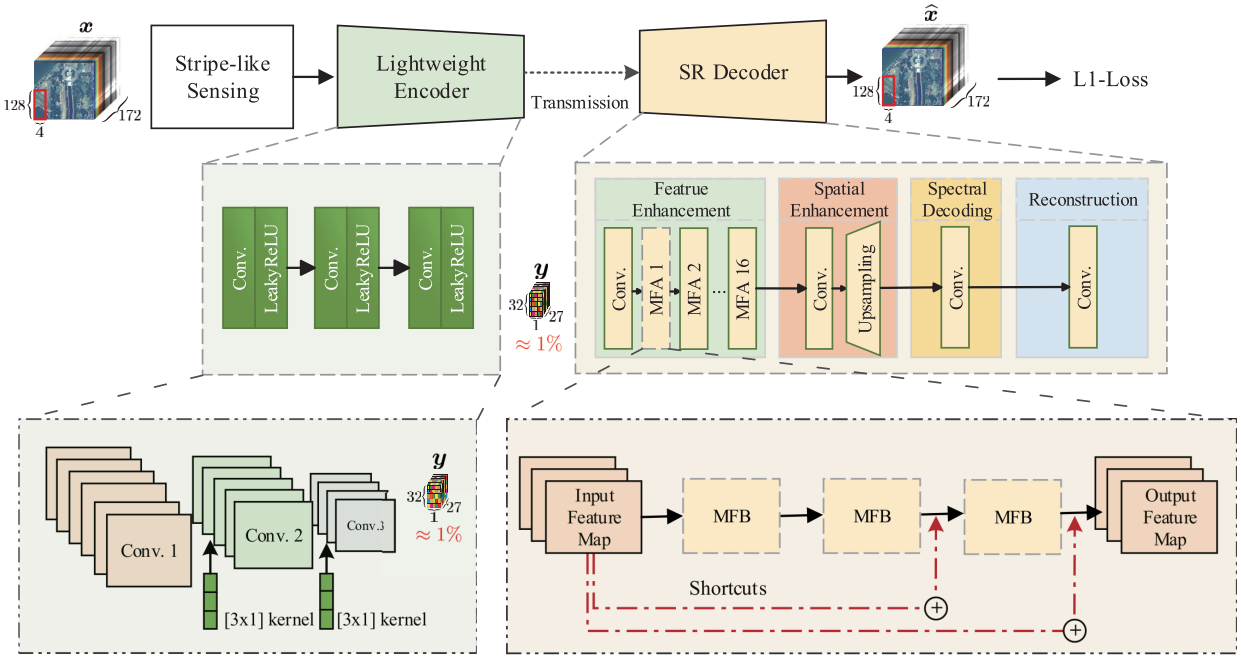


Fig. 3. Network architecture of the proposed DCSN for hyperspectral compressed sensing. (Left) Hyperspectral data $\boldsymbol{x}$ are compressively sensed to obtain the onboard acquisition $\boldsymbol{y}$, which is transmitted to the receiver on the ground station. (Right) Decoder will then upscale the compressed and noise-corrupted $\boldsymbol{y}$ to recover the decoded HSI $\widehat{\boldsymbol{x}}$.

2) Because the storage space of the miniaturized satellite is extremely limited, the unnecessary components should be eliminated as far as possible.
3) Because the computational power is also limited, the model complexity of the compressed sensing module used onboard the miniaturized satellite should be minimized.

In order to meet the abovementioned massive requirements, a lightweight encoder with a stripe-like data sensing strategy is proposed. Give a hyperspectral data $\boldsymbol{X} \in \mathbb{R}^{M\times L}$ to be sensed, where M and L are the numbers of spectral bands and the number of pixels, respectively. The stripe-like data are defined as

$$\boldsymbol{X}_i^s = \boldsymbol{R}\boldsymbol{X} \tag{1}$$

where $\boldsymbol{R}$ denotes a sensing operator used in the pushbroom system for capturing the ith stripe-like data, denoted as $\boldsymbol{X}_i^s \in \mathbb{R}^{M\times K}$ with $K \ll L$, where $K = r_1 r_2$ (see Fig. 2) is the number of pixels in a stripe. In most literature, the conventional CAE is designed for square-like image data and unable to process such stripe-like data due to the squared kernel size, such as 3×3 or 5×5.

The width and height (spatial resolution) of the output feature map can be calculated as $\lfloor (w+2p-k_s)/s \rfloor + 1$ and $\lfloor (h+2p-k_s)/s \rfloor + 1$, respectively, where w (respectively, h) is the width (respectively, height) of the input feature map, p is the number of pixels to be padded, k_s is the kernel size, and s is the convolutional stride used in the convolution operation. Using the 3×3 kernel for example, the width and height of the output feature map of the stripe-like data $\boldsymbol{X}_i^s \in \mathbb{R}^{M\times(r_1 r_2)}$ will be (r_1-2) and (r_2-2), respectively, given zero padding $p=0$ and unit stride $s=1$. It is obvious that the width w or height h of the $\boldsymbol{X}_i^s$ should be large than $k_s - 1$ to meet the minimal requirement of performing the convolutional operation for obtaining its feature map, restricting the minimal stripe size to be greater than the kernel size, for which it becomes difficult to stack multiple convolutional layers to obtain the hierarchical feature representation as the spatial size of the feature map will be reduced progressively.

One solution is to have a sufficiently large padding factor p, together with the minimum stride $s=1$, for keeping the size of the feature map invariant. However, padding operation is adopted to enlarge the spatial size of the input data without increasing any meaningful information. In other words, the calculations of those padded regions are ineffective. To maximize the utilization of the limited computational resource of the miniaturized satellite, we propose a nonsquare kernel size to obtain the feature maps of the stripe-like hyperspectral data $\boldsymbol{X}_i^s$. The network architecture of the proposed encoder is illustrated in Fig. 4. The encoder only stacked three convolutional layers consisting of one 3×3 and two 3×1 kernels with LeakyReLU activation function ρ, defined as

$$\rho(x) = \max(0,x) + n_s \times \min(0,x) \tag{2}$$

where n_s is the negative-side slope set as 0.2 in our experiments; comparing to ReLU, LeakyReLU avoids the well-known issue of dead neurons. Therefore, the compressed data $\boldsymbol{Y}$ can be represented as

$$\boldsymbol{Y}_i^s = e_3\big(e_2\big(e_1\big(\boldsymbol{X}_i^s, \boldsymbol{W_1}\big), \boldsymbol{W_2}\big), \boldsymbol{W_3}\big) \tag{3}$$

where function $e(\boldsymbol{X},\boldsymbol{W}) \equiv \rho(\boldsymbol{X}\otimes\boldsymbol{W})$ denotes the convolutional-activation operation, and $\boldsymbol{W} \in \mathbb{R}^{k_s^h \times k_s^w}$ is a nonsquare $k_s^h \times k_s^w$ convolutional kernel (filter). In this way, the compression ratio of the proposed encoder is flexible

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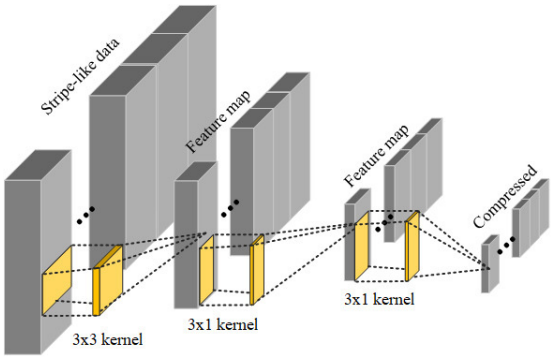


Fig. 4. Network architecture of the proposed lightweight encoder.

and easily adjustable by changing the stride value s, padding size p, and kernel size $k_s^h \times k_s^w$ to obtain the compressed hyperspectral data with different sampling rates. We will discuss how to choose these hyperparameters for different compression ratios in the experiments.

C. Super-Resolution Decoder

Accurately recovering the hyperspectral data from low-dimension and compressed signal, an effective SR technique is considered. Many DL-based SR techniques were proposed recently, beneficial from both CNN and GANs, thereby yielding a great improvement in the visual quality of the reconstructed image [19]–[21]. Those SR techniques are then modified to super-resolve the hyperspectral data (see [29], [31], [32], [36]). Nevertheless, all these methods just concentrate on upscaling the spatial resolution and completely ignore the enhancement of the spectral resolution. Note that separately performing SR for each HSI band would easily lead to distorted spectral signatures, seriously degrading the value of HSI. It is reported in recent literature [33, Section IV.C] that joint learning across all spectral bands is critical for preserving spectral properties in multiband SR. In this section, a highly accurate hyperspectral data SR technique is proposed for enlarging both spatial and spectral resolutions while decoding the compressed acquisition.

Our decoder comprises four parts to progressively restore the spatial information and spectral information from the noise-corrupted compressed data to achieve joint spatial and spectral resolution enhancement. Fig. 5 shows the flowchart of the proposed SR-aware decoder. First, an effective backbone is selected to enrich the feature representation for the compressed signal $\boldsymbol{Y}_i^s$. Next, we upscale the spatial resolution using transposed convolutional layers. Then, an extra convolutional layer is adopted to learn how to decode the spectral information from the output of the previous stage. Finally, the full spectral information can be super-resolved by the last convolutional layer.

Feature Representation Enhancement: The most important role of SR-aware CNN is the choice of the backbone networks. Recently, many advanced backbone network architectures were proposed, such as ResNet [30], DenseNet [37], and EfficientNet [38]. EfficientNet shows the best performance in image classification tasks. However, it was designed by the network architecture search (NAS) technique for image classification but not for SR. Instead, to have a better feature representation ability of the compressed hyperspectral data, we take the advantages of both ResNet and DenseNet to build the backbone network. Instead of putting such complicated network components on the encoder, we remain most parts of feature representation learning on the decoder to boost the efficiency of the communication between the miniaturized satellite and the ground station.

Inspired by [19], [20], we take the advantages from both DenseNet [37] and ResNet [30] to design our MFB for building the MFA unit, as shown in Fig. 6. Specifically, we stack three MFBs to form an MFA unit with two shortcut connections between MFBs, as illustrated in Fig. 3, where the MFB is further detailed in Fig. 6. Generally speaking, the multiscale feature is usually adopted to enhance the ability of feature representation for various computer vision tasks, such as object detection [24] and semantic segmentation [25]. It is well known that the features with a larger receptive field can be made by using a larger kernel size, but this also increases the number of the parameters and the memory requirement. A common way to reduce the number of the parameters is to stack several convolutional layers with a smaller kernel size, which is equivalent to a large one. For example, a feature map convoluted by a 5×5 kernel is equivalent to the feature map obtained by stacking two convolutional layers with 3×3 kernels. In our MFB, each feature map will be squeezed by a convolutional layer named squeeze layer, followed by concatenating it to the next feature map. As an example shown in Fig. 6, in the second layer (say, green and blue blocks), the input feature map (green and blue) will be convoluted by a 3×3 kernel and followed by concatenating them to the next layers (brown, blue, and green blocks), where the feature map indicated in brown is equivalent to have a larger receptive field as it has passed through two 3×3 kernels. Therefore, we achieve the multiscale feature aggregation and fusion module by the nature of the equivalence between the large receptive field-wise feature map with one larger kernel size and the several stacked convolution operations with smaller kernel size. Furthermore, to avoid the number of the channels increasing rapidly in the MFB, the squeeze layer is then adopted to reduce the number of the channels before the feature map concatenation. Assuming that the number of channels of the feature map $f(\boldsymbol{X}, \boldsymbol{W})$ is n_f and that of the squeezed feature map is g_f, the number of channels of the concatenated layer will be $2g_f$. Finally, we squeeze the final feature map to obtain the compact representation and add a shortcut connection between the input and the output feature maps.

Specifically, the gth output feature map $\boldsymbol{Y}_{R_g}$ is defined as

$$\boldsymbol{Y}_{R_g} = f_{R_g}\left(\boldsymbol{Y}_i^s, \boldsymbol{W}_{R_g}\right) \tag{4}$$

where f_{R_g} indicates the function of MFB and $\boldsymbol{W}_{R_g}$ is the set of the parameters in the MFB. In the proposed SR-aware decoder, we first enhance the ability of feature representation by stacking three MFBs. Since each output of the MFB has an identical shape, the shortcut connection is added between each MFB to further improve the ability of feature representation. In

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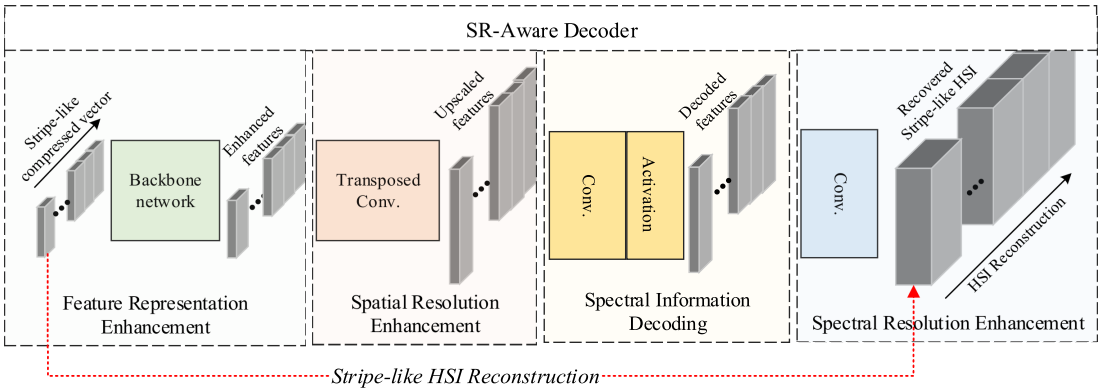


Fig. 5. Flowchart of the proposed SR-aware decoder for compressively sensed hyperspectral data.

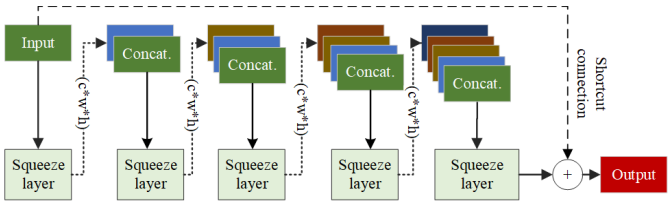


Fig. 6. Network architecture of the proposed MFB block.

this study, an MFA is made by stacking three MFBs, as shown in Fig. 3. Finally, we stack $k = 16$ MFAs to form the backbone network for our SR-aware decoder.

Spatial Resolution Enhancement: Afterward, the spatial resolution of the intermediate feature representation $\boldsymbol{Y}_R$ through our backbone network for the compressed signal $\boldsymbol{Y}_i^s$ can be enlarged by transposed convolution operation [20]. In practice, the spatial resolution can be upsampled by $2\times$ conventional interpolation (e.g., linear interpolation) and follow by concatenating a convolutional layer to learn how to accurately recover the fine-detailed feature map caused by the interpolation as follows:

$$\boldsymbol{Y}_{\text{up}} = \rho\left(f_{\text{up}}\left(\uparrow \boldsymbol{Y}_R, \boldsymbol{W}_{\text{up}}\right)\right) \tag{5}$$

where f_{up} is the convolutional layer, $\boldsymbol{W}_{\text{up}}$ is the weights of the function f_{up}, and $\uparrow \boldsymbol{Y}_R$ indicates the upsampled feature map generated by our backbone network.

Spectral Resolution Decoding and Enhancement: Finally, the spectral resolution is enlarged by two convolutional layers

$$\widehat{\boldsymbol{X}}_i^s = f_{\text{SPE}}\left(\rho\left(f_{\text{SPA}}\left(\boldsymbol{Y}_{\text{up}}, \boldsymbol{W}_{\text{SPA}}\right)\right), \boldsymbol{W}_{\text{SPE}}\right) \tag{6}$$

where $\boldsymbol{W}_{\text{SPA}}$ and $\boldsymbol{W}_{\text{SPE}}$, respectively, indicate the set of parameters for the spectral resolution enhancement functions f_{SPA} and f_{SPE}; the subscript "SPA" is to reflect the fact that size of $\boldsymbol{Y}_{\text{up}}$ remains the same as the one from the previous spatial SR layer, while "SPE" is to indicate SR along the spectral dimension. In practice, the number of channels of f_{SPA} is the same as that of $\boldsymbol{Y}_{\text{up}}$, while the number of channels of f_{SPE} is the same with the original hyperspectral data $\boldsymbol{X}_i^s$. Since there may exist beneficial information across bands of $\boldsymbol{Y}_{\text{up}}$ for recovering HR spectral result, an additional convolutional layer is used to learn latent feature representation for $\boldsymbol{Y}_{\text{up}}$. Afterward, the last convolutional layer is adopted to learn fine-detailed spectral information. Finally, a high-quality reconstructed hyperspectral data $\widehat{\boldsymbol{X}}_i^s$ can be obtained. Note that the reconstructed hyperspectral data are also stripe-like. Since current stripe-like data reconstruction is irrelevant to other stripes to be super-resolved, the SR process can be accelerated by parallel computing.

D. Optimization of DCSN

The criterion between the reconstructed hyperspectral data $\widehat{\boldsymbol{X}}_i^s(k)$ and the ground truth $\boldsymbol{X}_i^s(k)$ is ℓ_1 loss, which is defined as

$$L_{\ell_1} = \sum_{i=0}^{B}\sum_{k=0}^{V-1}\left\|\widehat{\boldsymbol{X}}_i^s(k) - \boldsymbol{X}_i^s(k)\right\|_1 \tag{7}$$

where B and $V = r_1 r_2$ are the batch size and the total number of data points in the stripe-like hyperspectral data, respectively. Based on backpropagation [39], the gradient $\bigtriangledown L_{\ell_1}$ calculated by derivation of our L_{ℓ_1} criterion (7) can be backpropagated to simultaneously update the parameters in the lightweight encoder and SR-aware decoder of the proposed DCSN.

E. Robust DCSN

Since the transmission between the satellite and the receiver on the ground may often be affected by noise, a robust model for encoding and decoding hyperspectral data $\boldsymbol{X}_i^s$ is desired. Inspired by denoising AE (DAE) in [26], the additive noise pattern can be used to corrupt the compressed acquisition, i.e., the input of DCSN decoder, to learn the robust DCSN model. In [26], the noise pattern is added to the input (i.e., hyperspectral data $\boldsymbol{X}_i^s$) and optimizes the model by backpropagation. Considering that our DCSN is an asynchronous AE architecture consisting of a lightweight encoder and a highly accurate decoder, it is hard to train an effectively robust encoder due to the oversimplified architecture. Instead, the noise is added to the encoded hyperspectral data $\boldsymbol{Y}_i^s$ to learn the robust decoder. Since the SR-aware decoder is more sophisticated and effective, it is more feasible to learn the robust DCSN model. To be precise, the noise-corrupted $\boldsymbol{Y}_i^s$ will be

$$\widetilde{\boldsymbol{Y}}_i^s = \boldsymbol{Y}_i^s + \mathbb{N}_r \tag{8}$$

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TABLE I
SPECIFICATIONS OF THE COLLECTED HSI DATA USED IN OUR EXPERIMENTS

INDEX	A	B	C	D	E	F
Site Name	Ft. Myers - Tampa /St. Petersburg, FL	Arizona Fire 1	Hama Havo 6, HI	Seney Refuge-ne9	Dewar Refill	Yellowstone NP 1B, WY
Country	America	America	America	America	America	America
Location	FL	AZ	HI	MI	WA	WY
Year	2010	2011	2007	2013	2008	2006
Data Type	Radiance	Radiance	Radiance	Radiance	Radiance	Radiance
GSD (m)	16.9	14.7	2.1	3.3	17.1	14.3
Spectral range (μm)	0.4–2.5	0.4–2.5	0.4–2.5	0.4–2.5	0.4-2.5	0.4-2.5
Data Size (pixels)	11340×904	5711×859	9672×940	5157×816	2447×742	11273×792
INDEX	G	H	I	J	K	L
Site Name	Lethbridge, Alberta	Amikougami Lake, Ontario	QC-Cut	Campbell River - Young Plantation Site, BC	HyspIRI_3_ line17	HyspIRI_3_ line40
Country	Canada	Canada	Canada	Canada	America	America
Location	AB	ON	QC	BC	NV, CA, OR	NV, CA
Year	2008	2008	2008	2008	2018	2018
Data Type	Radiance	Radiance	Radiance	Radiance	Reflectance	Reflectance
GSD (m)	16.1	16.9	16.6	16.2	14.9	13.6
Spectral range (μm)	0.4–2.5	0.4–2.5	0.4–2.5	0.4–2.5	0.4-2.5	0.4-2.5
Data Size (pixels)	3127×805	2567×763	3960×777	2542×829	42043×1280	46229×912

where $\mathbb{N}_r$ is zero-mean Gaussian noise generated based on a prespecified signal-to-noise (SNR) ratio $r = 10\log_{10}(p_{\text{signal}}/p_{\text{noise}})$ (in dB), in which p_{signal} and p_{noise} are the signal power and noise power, respectively.

With noised and compressed hyperspectral data, it is relatively hard to train a model for the proposed SR-aware decoder from scratch. Instead, the pure DCSN model is first trained under the noise-free hyperspectral data and follows by fine-tuning the parameters of the decoder based on the noised encoded signal $\widetilde{\boldsymbol{Y}}_i^s$. In this manner, our robust DCSN, termed RDCSN, is more effective in both transmission and SR in practice.

III. EXPERIMENTAL RESULTS

A. Experimental Settings

Dataset: We train the proposed DCSN using just small datasets, collected by standard HSI sensor known as Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [40]. These HSI data obtained from the AVIRIS Data Portal [41] contain 224 bands (with wavelengths ranging from 400 to 2500 nm), where low-quality bands (including bands 1–10, 104–116, 152–170, and 215–224) are removed [14], leading to a total of 172 bands used in this study. Besides, the negative value in our data sets is artificially set to zero so that it conforms to the nonnegative nature of the hyperspectral image. These data sets are acquired over California (CA), Oregon (OR), Nevada (NV), Florida (FL), Arizona (AZ), Hawaii (HI), Michigan (MI), Washington (WA), and Wyoming (WY) in America, as well as Alberta (AB), Ontario (ON), Quebec (QC), and British Columbia (BC) in Canada. Due to space limitations, we cannot show all the specifications of all the data sets. Instead, some representative ones are collectively introduced in Table I, where "Site Name" denotes the file name defined on the data portal [41], which is recorded here for explicitly indicating the data set information.

From these sites, a total of 2537 HSI sized of 256 × 256 × 172 are used in the experiment. Since data-driven approaches, such as DL-based schemes, rely on an assumption that the distributions of the test and the training samples should be similar to each other, we should verify the generalization ability of the proposed DCSN. Therefore, the collected HSI data sets are categorized into four subsets according to its visual appearance (land cover type), as summarized in the following:

1) 102 subimages for city areas, denoted as C-type;
2) 1870 subimages for mountain areas, denoted as M-type;
3) 272 subimages for vegetation areas (farm or grass), denoted as F-type;
4) 293 subimages for lake/coastline regions, denoted as L-type.

We randomly select 90% and 10% from each type of the data set as the training and test sets; specifically, the numbers of the subimages in the training set for C-, M-, F-, and L-types are 91, 1699, 249, and 261, respectively.

The proposed DCSN is trained on each type of training set and is evaluated on every type of test datasets, in order to verify the generalization property of the proposed method. In this section, all the averaged performance indices (SAM/PSNR/RMSE, defined next) are weighted averaged according to the proportions of the involved four-type test data sets. The union of all the four types of training sets is called the "All-type" training set, which is obviously isolated to each test set.

Network Hyperparameters: The learning rate during the training phase is 3e–4 with the AdamW optimizer [42] without decay. The batch size is 20, and the size of the stripe-like HSI input $\boldsymbol{X}_i^s$ is set as 128×4 pixels with 172 spectral bands. The total number of epochs of our DCSN is 20 000, and the training produce is terminated when the validation loss converges. During the training process, the standard data augmentation strategies (e.g., random flipping horizontally and/or vertically) are used to improve the generalization ability of the proposed DCSN architecture.

Performance Evaluation: To illustrate the superiority of DCSN, five state-of-the-art HCS methods are selected as

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TABLE II
SAMPLING RATE SETTING IN DCSN WITH RESPECT TO THE STRIPE-LIKE INPUT SIZE $128 \times 4 \times 172$

	Size of $\boldsymbol{Y}_i^s$	Exact Sampling Rate
$\approx 1\%$	$32 \times 1 \times 27$	0.981%
$\approx 5\%$	$64 \times 2 \times 32$	4.651%
$\approx 10\%$	$64 \times 2 \times 64$	9.302%
$\approx 15\%$	$64 \times 2 \times 103$	14.970%
$\approx 20\%$	$64 \times 2 \times 140$	20.340%

baselines, including SPACE [17], SpeCA [6], TenTV [12], LSS [15], and HyperCSI-LSS. As aforementioned, the default spectral library adopted in LSS yields weak HCS performances; as can be seen next, HyperCSI [14] is, hence, adopted to extract scene-adapted spectral library from all training samples to improve its performance, leading to HyperCSI-LSS. However, it would be hard to ensure the existence of such a complete library in the general scenario. We evaluate the performance of the proposed DCSN, as well as the five HCS methods, for sampling rates ranging from 1% to 20%, though the target sampling rate is the very small one (say, 1%). Since most HCS methods cannot precisely achieve an arbitrarily specified sampling rate due to the limited combination of valid hyperparameters, we tried to select the closest available one as an approximation instead. For example, given a stripe-like HSI sized of $128 \times 4 \times 172$ for our DCSN with a 1% sampling rate setting, the shape of the compressed HSI is $32 \times 1 \times 27$; therefore, the exact sampling rate (0.9811) is actually less than 1%. For each sampling rate under test, the corresponding size of the compressed acquisition $\boldsymbol{Y}_i^s$ obtained by our lightweight encoder in DCSN is given in Table II.

B. Quantitative and Qualitative Results

Mathematically, an HSI can be represented as a matrix, whose columns represent pixel vectors and whose rows represent spectral bands; this is commonly seen especially in geometry-based HSI analysis [5] and adopted here for rigorous definitions of the performance indices. Let $\widehat{\boldsymbol{X}}^{(m)}$ and $\boldsymbol{X}^{(m)}$ denote the mth rows/bands of the decoded/reconstructed HSI $\widehat{\boldsymbol{X}}$ and the ground-truth HSI $\boldsymbol{X}$, respectively. Then, the adopted performance indices are defined as follows.

1) Spectral angle mapper (SAM), given in degrees, is for measuring the spectral distortion

$$\text{SAM} = \frac{1}{L}\sum_{n=1}^{L} \arccos\left(\frac{(\widehat{\boldsymbol{x}}[n])^T \boldsymbol{x}[n]}{\|\widehat{\boldsymbol{x}}[n]\|_2 \cdot \|\boldsymbol{x}[n]\|_2}\right)$$

where $\widehat{\boldsymbol{x}}[n]$ denotes the nth column/pixel of $\widehat{\boldsymbol{X}}$. The smaller the value of SAM, the better the spectral reconstruction of $\widehat{\boldsymbol{X}}$.

2) The peak signal-to-noise ratio (PSNR), given in dB, is for measuring the spatial quality of the mth band

$$\text{PSNR}_m = 10\log_{10}\left(\frac{\max\left\{\widehat{x}_{mn}^2 \mid n \in \mathcal{I}_L\right\}}{\frac{1}{L}\|\widehat{\boldsymbol{X}}^{(m)} - \boldsymbol{X}^{(m)}\|_2^2}\right) \tag{9}$$

whose spectral average is then

$$\text{PSNR} = \frac{1}{M}\sum_{m=1}^{M} \text{PSNR}_m$$

where $\widehat{x}_{mn}$ denotes the nth entry in the vector $\widehat{\boldsymbol{X}}^{(m)}$, and $\mathcal{I}_L \triangleq \{1, \ldots, L\}$. The larger the value of PSNR, the higher the spatial quality of $\widehat{\boldsymbol{X}}$.

3) The root-mean-square error (RMSE) is for measuring the global quality

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{m=1}^{M} \text{RMSE}_m^2}$$

where

$$\text{RMSE}_m = \frac{1}{\sqrt{L}}\|\widehat{\boldsymbol{X}}^{(m)} - \boldsymbol{X}^{(m)}\|_2. \tag{10}$$

The smaller the value of RMSE, the higher the global quality of $\widehat{\boldsymbol{X}}$.

We first discuss how the cross-dataset affects the performance of the proposed DCSN. The DCSN model will be trained on each type of training sets and evaluated on every type of test set. For example, the evaluated results of *DCSN (C)* are based on our model trained on the C-type training set. Note that even the types of training and test sets are identical, the test HSIs are isolated to the training HSIs, as previously explained. In Fig. 7, the performance comparison among different types of the training set is conducted in terms of the SAM index, arguably the most important index for HSI quality assessment. Predictably, the performance corresponding to the same type of both training and test sets should be superior. As a result, as shown in Table III, the F-type test set has the best overall performance on average, alluding that the intrinsic features of F-type data are similar to that of other types. On the other hand, the best performance among different training sets is yielded by the M-type training set due to the simple fact of the largest number of available training HSI subimages. We also can observe that the performance gaps between each type of the test sets are closer to each other, implying that the DCSN model trained on the M-type training set is more stable and generalization due to a relatively large training set. Furthermore, we can see that the performance evaluated on the C-type test set is better than that on the M-type test set, showing the ability of the generalization of the proposed DCSN. Once the number of training subimages and their diversity can be augmented, the performance can be further improved. As we stated previously, the numbers of available C-, F-, and L-type HSI subimages are relatively small compared with that of M-type HSI subimages. Nevertheless, all of our DCSN models outperform the current state-of-the-art (i.e., SPACE), leading to a conclusion that the proposed DCSN is effective even under cross-dataset setting with small data (i.e., very limited number of training samples).

Next, the performance comparison is conducted between the proposed DCSN and the other state-of-the-art HCS schemes. Tables IV and V present the performance comparison between the proposed DCSN and other state-of-the-art methods for

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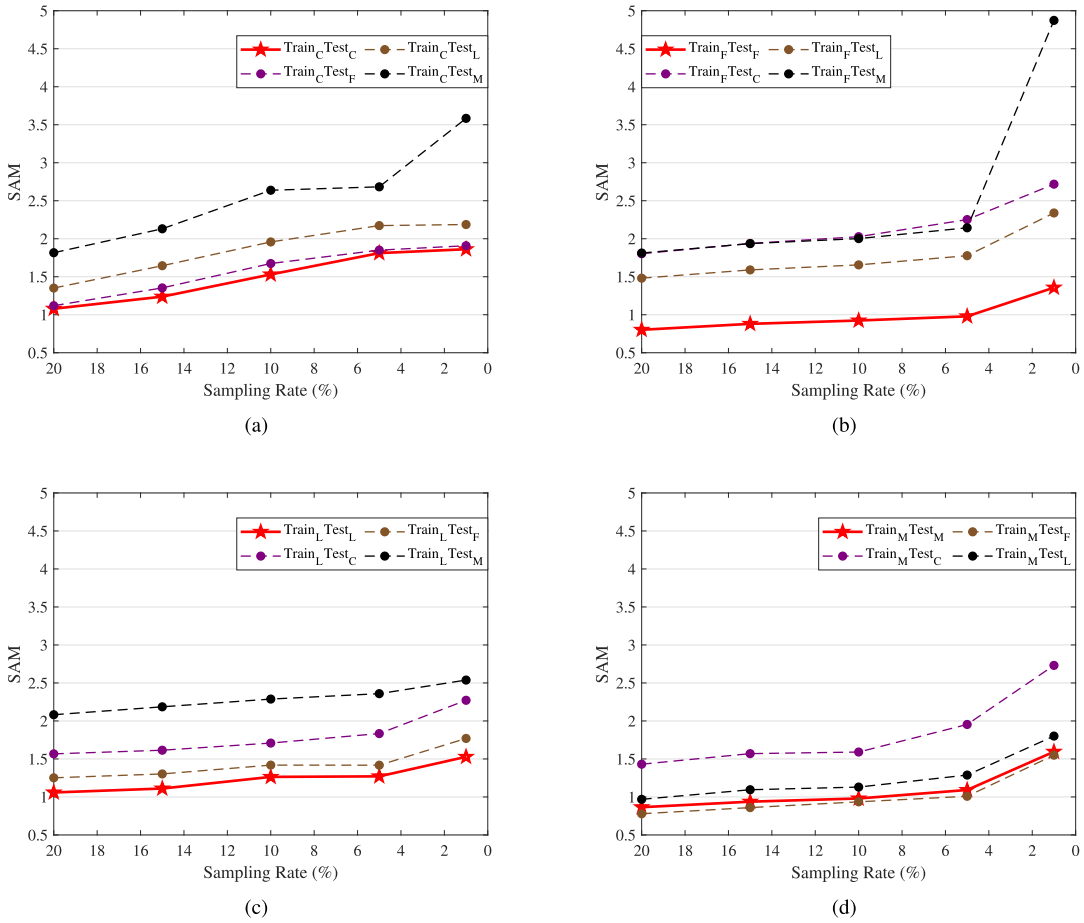


Fig. 7. Performance evaluation of the proposed DCSN for different sampling rates (from 1% to 20%) based on models trained on (a) C-type, (b) F-type, (c) L-type, and (d) M-type training sets.

TABLE III

PERFORMANCE COMPARISON OF THE PROPOSED DCSN FOR CROSS-DATASET SETTING UNDER SAMPLING RATE ≈ 1% IN TERMS OF PSNR/RMSE/SAM. AVERAGE PERFORMANCE ACROSS ALL THE FIVE VERSIONS OF DCSN IS ALSO DISPLAYED

Training \ Test Set	C-type	M-type	F-type	L-type	Averaged
	PSNR↑ / RMSE↓ / SAM↓				
DCSN (C)	34.551/62.437/1.862	30.260/50.947/3.584	34.267/19.361/1.908	33.463/24.162/2.187	31.280/44.798/3.152
DCSN (M)	33.188/78.269/2.731	33.752/30.652/1.595	35.327/18.978/1.550	34.567/27.795/1.801	33.988/31.343/1.671
DCSN (F)	32.834/77.508/2.718	30.074/68.014/4.873	35.750/17.657/1.357	33.137/29.138/2.339	31.166/58.318/4.089
DCSN (L)	33.666/70.175/2.272	31.806/39.403/2.538	34.541/20.117/1.770	34.972/22.456/1.528	32.585/36.671/2.314
Averaged	33.702/70.701/2.319	31.924/43.927/2.844	35.158/18.704/1.593	34.341/25.021/1.852	32.646/40.169/2.564
DCSN	**34.274/65.120/2.016**	**33.729/30.620/1.631**	**35.908/17.408/1.380**	**35.566/21.558/1.408**	**34.213/29.715/1.594**
SPACE [17] (≈ 1%)	24.129/613.661/7.207	29.161/140.415/3.743	29.674/64.151/3.121	27.727/209.757/4.446	28.783/164.342/3.983

sampling rate ≈1% and ≈5%. As shown in Table IV, the performance of proposed DCSN under sampling rate 1% is significantly better than that of other state-of-the-art methods. Moreover, the SAM index of the reconstructed HSI subimage by the proposed method is significantly better than that of other methods. It also implies that the proposed DCSN not only effectively super-resolves the spatial resolution but also accurately reconstructs the spectral information, indicating the effectiveness of the low-rank-driven [17] HCS network architecture design.

Similarly, our DCSN still achieves excellent performance under a sampling rate of 5% in most cases. SpeCA is a high-speed algorithm, which learns the data subspace directly from the RPs. Although its computation time is very fast, its reconstruction performance is not great when the sampling rate is very low. However, we can see that, when its compression ratio comes to 5%, its performance has been significantly improved because it acquires more information for subspace reconstruction. As one can observe, the low-rank modeling employed in the SPACE decoder yields very good spectral information reconstruction, as indicated by the values of the SAM index. TenTV spends quite an amount of computational time in reconstructing HSI, probably because its design requires processing the full HSI simultaneously even though

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TABLE IV

PERFORMANCE COMPARISON UNDER SAMPLING RATE ≈ 1% BETWEEN THE PROPOSED DCSN AND OTHER STATE-OF-THE-ART HCS METHODS IN TERMS OF PSNR/RMSE/SAM

Test Set / Method	C-type	M-type	F-type	L-type	Averaged
	PSNR↑ / RMSE↓ / SAM↓				
SPACE [17]	24.129/613.661/7.207	29.161/140.415/3.743	29.674/64.151/3.121	27.727/209.757/4.446	28.783/164.342/3.983
SpeCA [6]	9.299/784.867/42.863	15.377/234.735/21.510	11.701/407.530/33.036	14.024/225.772/22.006	14.555/275.827/23.687
TenTV [12]	20.208/570.255/26.247	18.533/260.221/22.972	20.401/248.994/18.714	18.824/314.248/25.523	18.831/280.816/23.055
LSS [15]	7.002/615.037/48.546	0.427/232.486/57.256	3.848/259.960/50.781	2.380/341.429/55.669	1.328/267.617/56.009
HyperCSI-LSS	25.078/278.263/8.704	26.146/51.421/4.907	25.943/82.299/5.732	25.897/83.626/5.779	26.043/69.294/5.281
DCSN (ours)	**34.274/65.120/2.016**	**33.729/30.620/1.631**	**35.908/17.408/1.380**	**35.566/21.558/1.408**	**34.213/29.715/1.594**

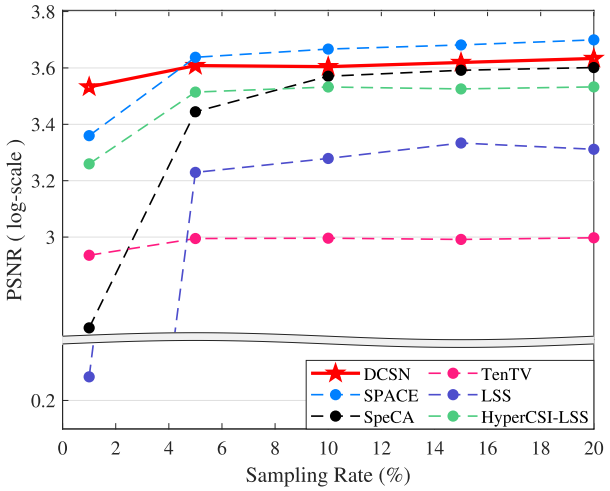


Fig. 8. PSNR comparison between the proposed DCSN and other state-of-the-art HCS methods.

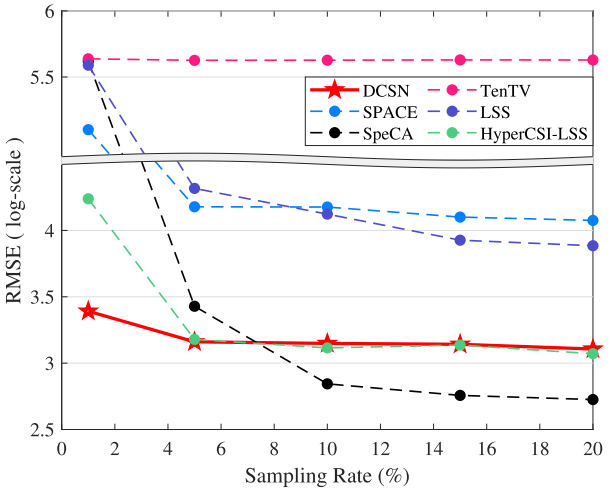


Fig. 9. RMSE comparison between the proposed DCSN and other state-of-the-art HCS methods.

it cannot handle such a challenging scenario, mainly due to the color distortion in the decoded image. LSS judiciously introduces hyperspectral unmixing into HCS but requires a very good library for decoding, making its performance not so strong among all peer methods. However, after using HyperCSI to compute a scene-adapted spectral library, the performance of LSS has been significantly improved.

In Fig. 8 and 9, as well as Fig. 1, we verify the performance of the proposed DCSN and other state-of-the-art methods in terms of PSNR, RMSE, and SAM. In the very low sampling rate setting (i.e., sampling rate ≈1%), the proposed DCSN trained on any type of data sets significantly outperforms the most advanced HCS method (i.e., SPACE [17]) in Table III. As the sampling rate increasing, the performance gap between the proposed DCSN and other state-of-the-art methods is insignificant. For a higher sampling rate (though less meaningful in the target application), the performance of SpeCA is outstanding in terms of RMSE, while SPACE is strong in terms of PSNR. Nevertheless, for all sampling rates under test, DCSN has the overall best performance, especially in terms of SAM. Note that the key value of HSI (comparing to other images) is its capability of material identification due to the densely populated spectral bands, and hence, SAM is regarded as the key performance index as it measures the spectral signature preservation. The remarkable performance of DCSN in terms of SAM also reveals the fact that the low-rank-driven-based joint learning [17] is indeed critical in high-fidelity spectral reconstruction, as reported in [33].

Qualitatively, RGB-visualization would let us understand more about distinguishing the good from the bad of each method. For each image, the 25th band, the 15th band, and the 6th band are used in the true-color composition. Fig. 10 presents the RGB-visualized results of the reconstructed HSI under the sampling rate of around 1%. It is clear that the reconstructed HSI by the proposed DCSN shows more visually pleasing results. From the reconstructed HSI shown in Fig. 10, the proposed DCSN is significantly better than other methods. For example, some of the HSI reconstructions obtained by other methods show the color-jitter artifacts, while the proposed DCSN can accurately recover the spectral information from the compressed acquisitions. DCSN achieves a very low sampling rate and, in the meanwhile, preserves visually pleasing reconstruction results, enabling a highly efficient transmission between the satellite and the receiver on the ground station.

C. Robustness Analysis

When transmitting the compressed acquisitions to the receiver, there are inevitable random noises, motivating us to study the robustness against noise for the HCS methods. Since our highly effective decoder is composed of several enhanced feature representation and distillation modules, the trained model itself would already be robust. Tables VI and VII present the objective quality comparison between the noisy

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TABLE V

PERFORMANCE COMPARISON UNDER SAMPLING RATE ≈ 5% BETWEEN THE PROPOSED DCSN AND OTHER STATE-OF-THE-ART METHODS IN TERMS OF PSNR/RMSE/SAM

Test Set / Method	C-type	M-type	F-type	L-type	Averaged
	PSNR↑ / RMSE↓ / SAM↓				
SPACE [17]	36.410/156.154/3.546	37.740/56.354/2.176	40.476/60.576/1.725	38.224/84.952/2.381	38.009/65.257/2.223
SpeCA [6]	31.889/83.437/3.544	31.225/25.837/2.965	31.028/24.432/3.434	31.901/44.076/2.642	31.329/30.837/2.994
TenTV [12]	22.238/562.588/25.804	19.630/256.872/22.569	21.603/246.708/18.560	19.928/312.549/25.155	21.864/498.172/25.013
LSS [15]	22.998/306.311/9.684	26.223/52.792/5.740	21.856/82.565/9.296	23.387/108.318/8.227	25.267/74.945/6.604
HyperCSI-LSS	34.779/68.808/3.661	33.228/21.398/2.612	34.497/17.363/2.355	34.497/27.440/2.632	33.594/24.023/2.638
DCSN (ours)	**37.167/50.829/1.230**	**36.412/24.398/1.042**	**38.867/13.551/0.880**	**38.051/17.114/0.953**	**36.907/23.589/1.023**

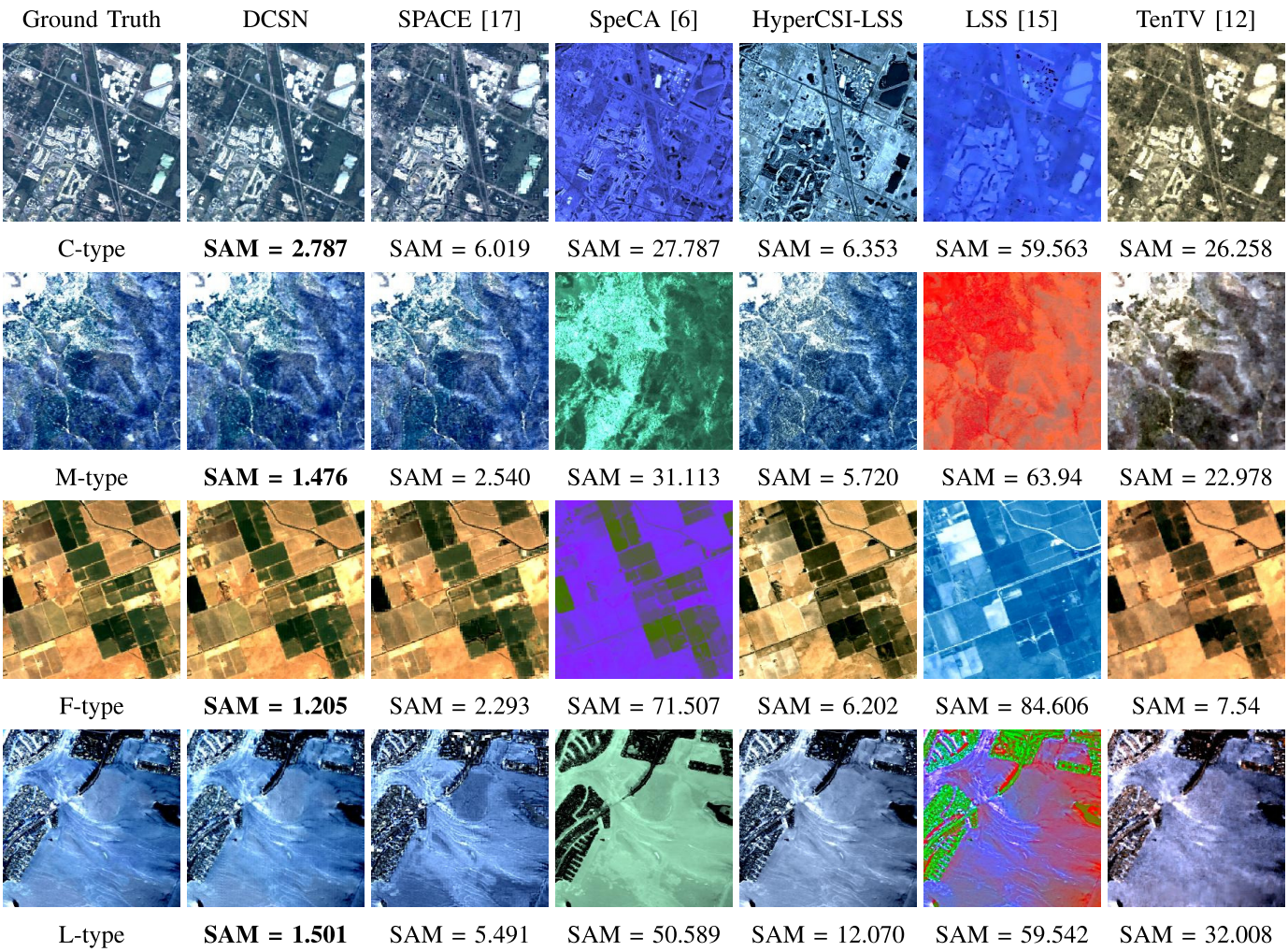


Fig. 10. RGB-visualized results of the (first column) ground truth and the reconstructed images under sampling rate ≈1% by using (second column) DCSN (trained on all-type), (third column) SPACE, (forth column) SpeCA, (fifth column) HyperCSI-LSS, (sixth column) LSS, and (seventh column) TenTV, for the four types of the test HSI samples, including (first row) C-type, (second row) M-type, (third row) F-type, and (forth row) L-type.

encoded signal and noise-free encoded signal in terms of PSNR, SAM, and RMSE. Unsurprisingly, the proposed DCSN also achieves good performance in noisy cases. Compared with SPACE [17] on the noise-free case, the performance evaluated on three metrics of SPACE is still lower than ours, leading to a fact that our DCSN is robust, and hence, it is a highly effective and efficient HCS architecture for HSI transmission and miniaturized satellite application.

Fig. 11 shows the SAM values comparison among different SNR settings of the proposed DCSN. To further demonstrate the advantage of the proposed DCSN, the SAM values of the prior art (i.e., SPACE [17]) with the noise-free case are added in Fig. 11 as the baseline. Our DCSN significantly outperforms SPACE [17]; even, the results of SPACE are measured in the noise-free case. In summary, our DCSN model is general and robust to noise. In this fashion, the reconstruction distortion due to imperfect communication between the satellite and the receiver can be minimized.

Furthermore, the aforementioned robust training strategy has also been considered. To further boost the robustness of

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TABLE VI

PERFORMANCE COMPARISON UNDER SAMPLING RATE ≈1% OF THE PROPOSED DCSN WITH SNR = 25 DB

Method \ Test Set	C-type	M-type	F-type	L-type	Averaged
	PSNR↑ / RMSE↓ / SAM↓				
SPACE [17]	26.541/186.318/7.357	24.854/56.425/5.754	25.463/67.489/5.609	25.194/74.333/5.919	25.037/65.945/5.837
SpeCA [6]	9.203/809.268/43.694	13.700/251.290/23.689	11.024/233.965/28.468	13.668/362.031/28.373	13.227/290.459/25.714
TenTV [12]	17.196/645.930/33.393	15.924/298.812/31.773	17.082/283.110/26.438	16.202/356.325/32.837	16.133/321.165/31.474
LSS [15]	3.668/1012.042/54.738	0.551/325.123/52.435	1.507/381.210/57.937	0.952/429.053/52.748	0.843/376.481/53.118
HyperCSI-LSS	22.180/366.104/12.385	22.758/75.110/6.743	21.935/140.315/8.799	22.608/137.825/8.698	22.631/103.412/7.468
DCSN (C)	32.872/72.512/2.498	28.975/49.412/3.802	32.114/25.509/2.485	31.302/30.709/2.811	29.775/45.639/3.480
DCSB (M)	32.399/84.178/3.012	32.270/35.510/2.016	34.018/22.779/1.841	32.996/31.691/2.192	32.544/36.018/2.069
DCSN (F)	32.144/82.226/2.997	29.242/70.795/5.164	34.343/21.231/1.711	31.856/32.328/2.685	30.225/61.322/4.394
DCSN (L)	33.085/74.407/2.535	30.771/42.779/2.853	33.449/23.352/2.043	33.375/26.499/1.902	31.490/40.164/2.631
DCSN	**33.283/72.457/2.408**	**32.049/36.275/2.095**	**34.302/22.220/1.737**	**33.436/27.274/1.893**	**32.512/35.375/2.048**

TABLE VII

PERFORMANCE COMPARISON UNDER SAMPLING RATE ≈1% OF THE PROPOSED DCSN WITH SNR = 35 DB

Method \ Test Set	C-type	M-type	F-type	L-type	Averaged
	PSNR↑ / RMSE↓ / SAM↓				
SPACE [17]	26.986/461.736/6.858	28.473/60.762/3.803	29.418/46.562/3.426	26.923/164.183/4.836	28.286/91.959/4.048
SpeCA [6]	9.167/788.894/43.005	15.011/236.704/21.793	11.541/220.503/27.535	14.826/337.435/26.979	14.378/274.362/24.035
TenTV [12]	19.596/576.071/26.875	17.959/263.349/24.021	19.349/252.176/19.694	18.384/317.059/26.282	18.227/284.031/24.039
LSS [15]	3.820/903.230/53.220	0.247/321.352/52.041	1.314/382.490/57.541	0.663/426.599/52.825	0.573/368.503/52.735
HyperCSI-LSS	23.077/360.758/12.081	25.033/63.357/6.119	23.954/122.709/7.620	25.136/114.909/7.572	24.851/89.880/6.738
DCSN (C)	33.495/66.958/2.185	29.986/45.600/3.444	33.291/21.426/2.164	32.569/26.346/2.467	30.818/41.646/3.129
DCSB (M)	33.105/78.896/2.762	33.557/31.212/1.651	35.176/19.377/1.585	34.360/28.222/1.851	33.802/31.873/1.723
DCSN (F)	32.761/77.996/2.749	29.971/68.339/4.908	35.576/18.044/1.400	32.971/29.460/2.379	31.050/58.657/4.126
DCSN (L)	33.609/70.630/2.302	31.682/39.789/2.574	34.407/20.452/1.801	34.773/22.890/1.576	32.453/37.062/2.352
DCSN	**34.165/65.859/2.063**	**33.511/31.263/1.692**	**35.721/17.885/1.424**	**35.277/22.138/1.472**	**33.994/30.338/1.654**

TABLE VIII

PERFORMANCE COMPARISON UNDER SAMPLING RATE ≈1% BETWEEN THE PROPOSED RDCSN AND OTHER STATE-OF-THE-ART HCS METHODS

Method \ SNR	25dB	30dB	35dB	40dB	45dB
	PSNR↑ / RMSE↓ / SAM↓				
SPACE [17]	25.037/65.945/5.837	27.375/58.923/4.426	28.286/91.959/4.048	28.840/111.435/3.832	28.985/163.000/3.893
SpeCA [6]	13.227/290.459/25.714	13.930/278.689/24.485	14.378/274.362/24.035	14.573/272.809/23.879	14.647/272.240/23.827
TenTV [12]	16.133/321.165/31.474	17.423/293.039/26.266	18.227/284.031/24.039	18.763/280.927/23.121	18.754/281.119/23.146
LSS [15]	0.843/376.481/53.118	0.673/371.146/52.884	0.573/368.503/52.735	0.629/367.350/52.706	0.589/366.480/52.706
HyperCSI-LSS	22.631/103.412/7.468	24.062/93.957/6.938	24.851/89.880/6.738	25.065/87.853/6.660	25.224/86.677/6.667
DCSN	32.512/35.375/2.048	33.573/31.633/1.764	33.994/30.338/1.654	34.141/29.914/1.614	34.191/29.778/1.600
RDCSN	**33.740/31.006/1.699**	**34.178/29.922/1.589**	**34.337/29.567/1.549**	**34.387/29.456/1.536**	**34.405/29.417/1.532**

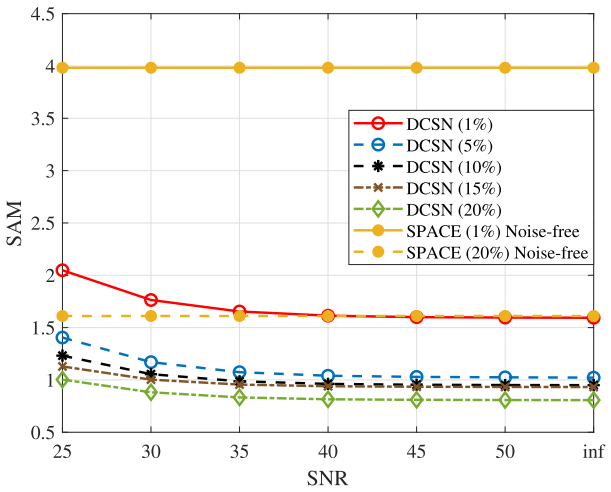


Fig. 11. Objective quality comparison across different SNR settings of the proposed DCSN.

the proposed DCSN, we fine-tune the pretrained models (see Section II-E) to obtain the robust DCSN models. Table VIII presents the robustness comparison between the proposed method and other methods in terms of SAM, PSNR, and RMSE, in which the performance is evaluated in all types of validation sets. The robust version of DCSN (i.e., RDCSN) is better than DCSN, especially in a scenario with significant noise. In contrast, most of the other optimization-based methods are less robust against noise when reconstructing the HSI information from the noised and compressed HSI. We show that the proposed DCSN and RDCSN do achieve excellent performance under noisy cases. Though SpeCA has good performance under a higher sampling rate (say, larger than 10%), its performance in noisy cases becomes poor, as the information for basis reconstruction is corrupted. It seems that SPACE achieves the best robustness among the peer methods, probably because it only applies the averaging-operators in the coding stage that has an effect of mitigating the noise. All in all, it is suggested that the RDCSN can be adopted for the real-world noisy transmission environment.

D. Complexity Analysis

The floating-point operations per second (FLOPS) of the proposed DCSN model is 0.866 Giga-FLOPs (GFLOPs) for a

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TABLE IX
COMPUTATIONAL TIME (IN SECONDS) OF HCS METHODS FOR DIFFERENT SAMPLING RATES (IN PERCENTAGE)

	DCSN-GPU	DCSN-CPU	DCSN-E-GPU	DCSN-E-CPU	SpeCA [6]	LSS [15]	HyperCSI-LSS	TenTV [12]	SPACE [17]
$\approx 1\%$	**0.0094**	0.43	**0.0021**	0.023	0.054	7.08	315.464	1274.11	16.59
$\approx 5\%$	**0.0121**	0.57	**0.0036**	0.022	0.090	2.66	480.621	1318.74	11.47
$\approx 10\%$	**0.0164**	0.66	**0.0052**	0.024	0.239	1.01	455.487	1400.35	13.00
$\approx 15\%$	**0.0223**	0.97	**0.0066**	0.028	0.418	1.02	592.122	1602.54	15.11
$\approx 20\%$	**0.0274**	1.16	**0.0098**	0.028	0.641	0.90	486.384	1554.92	19.34

stripe-like hyperspectral data sized of $256 \times 4 \times 172$, while the FLOP of the encoder part is 0.0288 GFLOPS only, in which the computational and space complexity is significantly less than decoder, satisfying the requirement of very lightweight hardware in the miniaturized satellite applications. In this fashion, several AI edge computing chips can be used to implement our algorithm in the miniaturized satellite. For instance, an Intel edge AI chip (i.e., Myriad X VPU) can be used on the satellite for data compression and transmission. According to the specifications of Intel VPU, it provides nearly 100 GFLOPS under 1-W power-consuming for inference, which is significantly larger than the requirement of encoder. Furthermore, the number of the parameters of our encoder is 0.228M only, making a minimized requirement of the memory storage. As a result, the proposed DCSN is efficient for HCS on miniaturized satellites.

We also tested the averaged run-time complexity of the encoding/decoding stages of the proposed DCSN, as shown in Table IX. The run time of DCSN excludes the training time by following the vast of DL literature, where most models are pretrained. Similarly, our DCSN can be pretrained before it is launched onboard the satellite for the HCS application. Note also that, in our target application, the weights of DCSN are well-trained (offline mode), as well supported by the experiments, meaning that it is unnecessary to retrain the entire model in every transmission. The run time is measured in a personal computer equipped with Intel i7-9900k CPU with 64-GB system memory and NVIDIA GTX-2080Ti GPU and evaluated on both GPU and CPU for observing the effect of various devices. The run time is calculated by averaging that of 30 hyperspectral data with a size of $256 \times 256 \times 172$. As a result, our DCSN achieves the fastest reconstruction speed without any code optimization, including encoding and decoding of hyperspectral data. Furthermore, our DCSN achieves a very fast encoding process (say, less than 0.03 s) on CPU only device, implying that our DCSN is suitable for deploying on any devices when the computational resources are limited (e.g., miniaturized satellite).

We also evaluate the run time of the encoder only of our DCSN, termed DCSN-E, to verify how efficient the compressed sensing process is in the miniaturized satellite. The run time of the DCSN-E is just around 2.1 ms for the 1% sampling rate, which is significantly less than that of other state-of-the-art HCS methods. Since the proposed lightweight encoder of the DCSN is much faster than the decoder, our encoder only requires extremely low power consumption (about 1 W, VPU as an example), leading to a high-quality and efficient HSI compressed sensing technique suggested to be implemented onboard the miniaturized satellite.

TABLE X
SETTINGS OF THE PROPOSED BASELINE MODELS

	Pushbroom	MFA
Baseline-I		
Baseline-II	$\surd$	
Baseline-III		$\surd$
DCSN	$\surd$	$\surd$

E. Ablation Study

The proposed DCSN is based on the lightweight encoder and MFA-based decoder. Three baseline models are proposed in this section to verify the effectiveness of each component of the proposed DCSN, as shown in Table X. There are nine convolutional layers to form the encoder for Baseline-I and Baseline-III. We also replace the proposed MFA with traditional ResBlock [30] to form the decoder for Baseline-I and Baseline-II. In the decoder of the Baseline-I and Baseline-II, we stack 64 ResBlocks to make its number of the parameters approximately given by that of our DCSN. Since our DCSN is designed for sensing the stripe-like hyperspectral data, it is necessary to verify the training efficiency for the complete HSI data (e.g., $256 \times 256 \times 172$ hyperspectral data). Therefore, we also train the proposed Baseline-I and Baseline-III for full HSI data instead. All the hyperparameters of the baseline models are identical to that of our DCSN.

Table XI presents the effectiveness and efficiency comparison between the proposed DCSN and other baseline models. The performance index (i.e., PSNR, RMSE, and SAM) and the run-time are measured for recovering a full HSI subimage sized $256 \times 256 \times 172$. Note that the FLOPS of Baseline-II and DCSN (pushbroom system) is measured for stripe-like hyperspectral data, while that of Baseline-I and Baseline-III is calculated for a square-like one. All the models are trained based on all types of the training set and evaluated on all types of test sets. When the proposed MFA is removed (i.e., Baseline-I and Baseline-II), the performance indices, such as SAM, PSNR, and RMSE, are decreased. Besides, the efficiency of the proposed lightweight encoder with a pushbroom scanning system can significantly reduce the FLOPs from $3.953G$ to $0.0288G$, while only two performance indices (PSNR and RMSE) are slightly decreased. It is interesting that the SAM index of the lightweight encoder with a pushbroom system is better than the complex encoder with a traditional sensing strategy (i.e., nonstripe-like HSI data). Since the stripe-like data slightly lack spatial information, the model is relatively hard to learn how to enhance the spatial resolution. However, such a pushbroom scanning strategy will

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TABLE XI
ABLATION STUDY UNDER SAMPLING RATE ≈1% BETWEEN THE PROPOSED DCSN AND ITS BASELINE MODELS

	SAM	PSNR	RMSE	FLOPS	FLOPS-E	#Parameters	#Parameters-E	Run-time	Run-time-E
Baseline-I	2.674	34.158 dB	30.248	59.501G	3.953G	5.270M	0.369M	0.293 sec.	0.081 sec.
Baseline-II	2.173	33.509 dB	32.005	**0.431G**	0.0288G	**5.129M**	0.0228M	**0.0093** sec.	**0.0020 sec.**
Baseline-III	2.009	**35.761** dB	**21.165**	57.529G	3.953G	12.091M	0.369M	0.434 sec.	0.078 sec.
DCSN	**1.594**	34.214 dB	29.716	0.866G	**0.0288G**	11.951M	**0.0228M**	0.0094 sec.	0.0021 sec.
SPACE [17]	3.983	28.783 dB	164.342					16.59 sec.	0.231 sec

tend to super-resolve spectral resolution, leading to the fact that the SAM index of Baseline-II is better than Baseline-I. Although it is reasonable to increase the depth of the encoder to improve the ability of the feature representation, the time and space complexity of the encoder will be increased as well. In this way, it is not easy to implement the real-time CS application on the miniaturized satellite. In summary, the proposed DCSN has the best tradeoff between the performance and the time-complexity, making the real-time compressed sensing application possible for the miniaturized satellite.

F. Limitation and Solution

The underlying assumption of any data-driven approach (e.g., DL) is to have similar data distribution between the test and training sets. The performance during the testing phase will be degraded when the nature of the test data is significantly different from that of the training data. For example, each band of the hyperspectral data corresponds to a specified wavelength or physical characteristic. Once the test data have quite different physical structures from training data, it is expected that the performance of the proposed DCSN is restricted. As we aim to develop a computationally cheap and lightweight HCS device on a miniaturized satellite, it is reasonable to assume that a miniaturized satellite would acquire a similar type of hyperspectral data during its tenure track. In other words, the underlying assumption of such a data-driven approach should hold in our application scenarios.

In case that acquiring a new type of hyperspectral data (e.g., acquired by a different type of sensor) is needed, we also propose a learning strategy called one-shot transfer learning (OTL). First, a complete DCSN model is trained on the original hyperspectral data set used in the previous experiments. Afterward, two new types of sensors are used to verify the performance of the proposed OTL strategy. The first data set was taken over Los Angeles, CA, USA, acquired by the Hyperion sensor equipped on NASA Earth Observing-1 (EO-1) satellite [43]. The second data set was taken over Washington DC Mall, USA, acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor [44]. The new sensor samples are then established upon these two data sets containing 20 subimages. We randomly pick up one HSI subimage per sensor as the training set, as well as randomly sample one HSI subimage per type in the original data set, resulting in 6 subimages in total for model fine-tuning via a lower learning rate $1e-6$ with 2000 epochs. The rest of Hyperion/HYDICE subimages are then used to form the test set, where a total of 18 HSI subimages are available for verifying the effectiveness of the proposed OTL-enhanced DCSN (OTL-DCSN) under the cross-sensor setting. Table XII shows the performance of OTL-DCSN at ≈1% sampling rate on the Hyperion/HYDICE test set. Since the parameters have been well-trained on the original data set, it is unnecessary to learn a model for new types of hyperspectral data from scratch, as well supported by the outstanding performance of the proposed OTL-DCSN.

TABLE XII
PERFORMANCE EVALUATION OF THE PROPOSED OTL STRATEGY ON THE HYPERION/HYDICE TEST SET

Training \ Testing	Performance Index
	PSNR↑ / RMSE↓ / SAM↓
DCSN	24.362 / 636.326 / 12.277
OTL-DCSN	**29.119 / 253.281 / 4.879**
SPACE	24.247 / 730.841 / 7.309

IV. CONCLUSION

We developed a new HCS deep neural network particularly customized for miniaturized hyperspectral satellite applications. In the proposed lightweight encoder, simplified network architecture with nonsquare filters has been proposed to efficiently compress the HSI within milliseconds. Besides, we also proposed a highly effective decoder network to progressively decoding the spatial and spectral resolutions from the compressed acquisitions received from the satellite. In our decoder, the MFB and MFA units have been proposed to fully discover and fuse the multiscale feature information, for enhancing the feature representation ability, ensuring effective spatial information reconstruction. Extra convolutional layers were then adopted at the end of the decoder to upsample the spectral resolution. This ordering of layers greatly reduces the number of parameters in the decoder, as inspired by the low-rank modeling of HSI achieving recent advances in HCS [17]. The proposed highly efficient encoder and low-rank-driven decoder are trained in an end-to-end [35] manner to obtain a joint HCS/SR architecture for joint spectral/spatial compression and enhancement. Furthermore, a training strategy for robust model learning is also proposed and verified to be effective. Experimental results show the superiority of the proposed DCSN over benchmark HCS methods, especially at a very low sampling rate.

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